

CoinGecko API — Reference

API documentation for the **CoinPulse — Crypto Tracker** project. List coins, show prices and coin details.

Base URL	<code>https://api.coingecko.com/api/v3</code>
Protocol	HTTPS
Auth	None (free "Demo" tier)
Format	JSON
Methods used	<code>GET</code>

Authentication & Headers

No key required for the free public tier. (An optional demo key can be sent as `x-cg-demo-api-key` for higher limits, but it's not needed here.)

Endpoint 1 — Coin markets (list)

Used by the **Home** (top movers) and **Markets** table screens. Returns an **array**, already sorted, with everything you need to render rows.

```
GET /coins/markets?vs_currency=usd&order=market_cap_desc&per_page=20
```

Query param	Required	Description
<code>vs_currency</code>	yes	Fiat to price in, e.g. <code>usd</code>
<code>order</code>	no	Sort, e.g. <code>market_cap_desc</code>
<code>per_page</code>	no	How many coins to return (1–250), e.g. <code>20</code>
<code>price_change_percentage</code>	no	e.g. <code>24h</code> for extra windows

Response **200 OK** (one element shown):

```
[
  {
    "id": "bitcoin",
    "symbol": "btc",
    "name": "Bitcoin",
    "image": "https://assets.coingecko.com/coins/images/1/large/bitcoin.png",
    "current_price": 62058,
    "market_cap": 1244496179202,
    "market_cap_rank": 1,
    "total_volume": 28636096610,
    "price_change_percentage_24h": 0.91666,
    "circulating_supply": 20051553
  }
]
```

Fields (per coin)

Field	Type	Description
<code>id</code>	string	Unique id, e.g. <code>bitcoin</code> — used for the detail call
<code>symbol</code>	string	Ticker, e.g. <code>btc</code> (lowercase)
<code>name</code>	string	Display name
<code>image</code>	string (URL)	Coin logo
<code>current_price</code>	number	Price in <code>vs_currency</code>
<code>market_cap</code>	number	Market capitalisation
<code>market_cap_rank</code>	number	Rank (#1 = largest)
<code>total_volume</code>	number	24h trading volume
<code>price_change_percentage_24h</code>	number	24h change (can be negative)
<code>circulating_supply</code>	number	Coins in circulation

Formatting note: prices/caps are raw numbers — format them client-side: `n.toLocaleString("en-US", { style:"currency", currency:"USD" })`. Colour `price_change_percentage_24h` green when `>= 0`, red when `< 0`.

Endpoint 2 — Coin detail

Used by the **Coin Detail** screen. Use the `id` from the markets list.

```
GET /coins/{id}
```

Example: `https://api.coingecko.com/api/v3/coins/bitcoin`

Returns a large object; useful paths: | Path | Description | |---|---| | `name` , `symbol` | Names | | `image.large` | Logo URL | | `market_data.current_price.usd` | Price | | `market_data.price_change_percentage_24h` | 24h change | | `market_data.market_cap.usd` | Market cap | | `market_data.total_volume.usd` | Volume | | `market_data.circulating_supply` | Supply | | `market_data.ath.usd` | All-time high | | `description.en` | HTML description text |

Endpoint 3 — Search (optional)

```
GET /search?query=bitcoin → results in `.coins[]` with id, name, symbol, thumb
```

Status & errors

Code	Meaning
<code>200 OK</code>	Success
<code>404 Not Found</code>	Unknown coin id
<code>429 Too Many Requests</code>	Rate limited — slow down / cache

Rate limiting

Free tier ≈ **5–15 calls/minute** per IP. For the markets table, make **one** `/coins/markets` call and render all rows from that array — don't call per coin.

Example requests

```
curl "https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd&order=market_cap_desc&per_page=10"
curl "https://api.coingecko.com/api/v3/coins/bitcoin"
```

```
const res = await fetch("https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd&order=market_cap_desc&per_page=10");
if (!res.ok) throw new Error(`HTTP ${res.status}`); // handle 429
const coins = await res.json(); // array – render the table
```

Screen → endpoint map

Screen	Endpoint
Home (top movers)	<code>/coins/markets</code> (slice the first few)
Markets table	<code>/coins/markets</code> (render all rows)
Coin Detail	<code>/coins/{id}</code>